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RESEARCH-ARTICLE

Vibration-Augmented Buttons: Information Transmission Capacity and Application to Interaction Design

CHAEYONG PARK, Pohang University of Science and Technology, Pohang, Kyongsangbuk-do, South Korea

JEONGWOO KIM, Pohang University of Science and Technology, Pohang, Kyongsangbuk-do, South Korea

DONG-GEUN KIM, Pohang University of Science and Technology, Pohang, Kyongsangbuk-do, South Korea

SEUNGJAE OH, Pohang University of Science and Technology, Pohang, Kyongsangbuk-do, South Korea

SEUNGMOON CHOI, Pohang University of Science and Technology, Pohang, Kyongsangbuk-do, South Korea

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Chaeyong Park
pcy8201@postech.ac.kr
Interaction Lab, POSTECH
Pohang, Republic of Korea

Jeongwoo Kim
jwkim0417@postech.ac.kr
Interaction Lab, POSTECH
Pohang, Republic of Korea

Dong-Geun Kim
dgkim94@postech.ac.kr
Interaction Lab, POSTECH
Pohang, Republic of Korea

Seungjae Oh
oreo329@postech.ac.kr
Interaction Lab, POSTECH
Pohang, Republic of Korea

Seungmoon Choi
choism@postech.ac.kr
Interaction Lab, POSTECH
Pohang, Republic of Korea

ABSTRACT

One can embed a vibration actuator to a physical button and augment the physical button's original kinesthetic response with a programmable vibration generated by the actuator. Such vibration-augmented buttons inherit the advantages of both physical and virtual buttons. This paper reports the information transmission capacity of vibration-augmented buttons. It was obtained by conducting a series of absolute identification experiments while increasing the number of augmented buttons. The information transmission capacity found was 2.6 bits, and vibration-augmented and physical buttons showed similar abilities in rendering easily recognizable haptic responses. In addition, we showcase a VR text entry application that utilizes vibration-augmented buttons. Our method provides several error messages to the user during text entry using a VR controller that includes an augmented button. We validate that the variable haptic feedback improves task performance, cognitive workload, and user experience for a transcription task.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *Haptic devices*; User studies.

KEYWORDS

Button, Vibrotactile, Haptics, Augmentation, Information Theory

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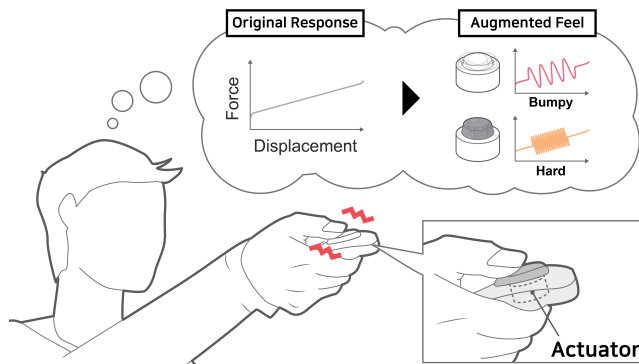


Figure 1: Concept of vibration-augmented buttons. A vibration actuator is embedded in a physical button. Upon detecting a keypress, a vibration is generated by the actuator and added to the original kinesthetic response of the physical button. As a result, the user perceives both kinesthetic (force and displacement) and tactile cues (vibration).

1 INTRODUCTION

Buttons, either physical and virtual, are a vital component in most input interfaces. Physical buttons provide high physical affordance and natural haptic feedback. Virtual buttons, which are adaptive and programmable, have replaced physical buttons in some cases, e.g., touchscreens. Recently, *physical buttons augmented with vibration feedback*, which inherit the advantages of both physical and virtual buttons, have emerged as a new alternative [35, 36, 46]; see Figure 1. This simple technology affords a variety of “feels” to a single physical button [36].

Ogawa et al. [35] first presented the idea of vibration-augmented buttons with a compact prototype. Shim and Lee [46] applied the idea to a gamepad for demonstration. Park et al. [36] expanded the idea by designing many augmented buttons and quantifying their perceptual effects. The authors estimated the pairwise perceptual distances between 7 physical and 40 augmented buttons and then constructed a 2D perceptual space¹. Two adjective pairs, *sharp-blunt* and *hard-soft*, were found to be the best perceptual dimensions².

¹An n -dimensional Euclidean space visualizing the intrinsic metric relationships between the stimuli of interest [58].

²Linguistically meaningful axes spanning the perceptual space.

Inspired by these previous studies, this paper proceeds to finding effective design methods for vibration-augmented buttons. To this end, we adopt the information-theoretic framework [50, 52] and quantify the information transmission capacity of vibration-augmented buttons. Given the number N of augmented buttons, we choose the most distinguishable augmented buttons from the perceptual space presented by Park et al. [36]. Then, we conduct an absolute identification experiment that estimates the information transfer (IT) of the selected augmented buttons. We repeat the above procedure while increasing N until the IT estimates are saturated. Consequently, we obtain the maximum amount of information transmitted from the source to the user through vibration-augmented buttons, that is, the *human channel capacity* of vibration-augmented buttons. This perceptual analysis based on the information theory allows us to determine the maximum number of augmented buttons that users can recognize accurately without errors. Section 3, 4, and 5 report the detailed procedure and results. Section 5 is about the IT of only physical buttons included for comparison.

We also showcase an example in which vibration-augmented buttons can be utilized. We apply the findings of the information-theoretic analysis to improving the traditional pointing-based text entry method for virtual reality. A user wearing a head-mounted display (HMD) operates a handheld controller equipped with a vibration-augmented button. The user points to a letter on a visual keyboard and enters it by pressing the augmented button. In our use scenario, the VR system provides several haptic messages to the user through the augmented button for assistance. By a user study adopting a transcription task, we demonstrate that our text entry method improves task performance, cognitive workload, and user experience. Further details are described in Section 6 and 7.

The main contributions of this work are (1) a thorough analysis of the information transmission capacity of vibration-augmented buttons and (2) a successful demonstration of an interactive application using augmented buttons.

2 RELATED WORK

2.1 Physical, Virtual, and Augmented Buttons

A physical button presents a unique feel determined by its size, shape, travel distance, and force-displacement curve [1, 54]. For example, Liu et al. [30] showed that the smoothness and travel distance of physical buttons are crucial factors in classifying the button sensations.

In contrast, the haptic response of virtual buttons³ can be varied [13]. Researchers have been keen to find effective design heuristics for virtual buttons. For example, Park et al. [39] designed 72 virtual buttons using different vibration patterns mimicking the acceleration profiles of real button clicks. Kildal [19] introduced vibration patterns that elicit a sensation of vertical movement when pressing a virtual button (called a compliance illusion). Kim and Lee [22] further improved the realism of virtual button sensations by referring to the force-displacement curve of a physical button. Sadia et al. [43] applied a data-driven approach, which mimics the input-output response measured from a physical system of interest, to virtual button design. Such efforts contributed to the realistic tactile key-click feedback available in recent commercial mobile

devices [2, 3]. However, virtual buttons still lack other informative tactile feedback, e.g., when the finger rests on a key or moves across keys; cf. TapBoard by Kim et al. [23].

Virtual buttons can improve usability in many tasks. For example, tactile feedback on a virtual keyboard displayed on a touchscreen enhances the typing performance [6, 16, 21]. Jeon et al. [18] explored the design space of user-adaptive key click vibrations and demonstrated that they could reduce the workload of typing. Kung et al. [25] designed perceptually different button sets and associated them with different regions of a QWERTY virtual keyboard to help users to locate the right keys.

As for the perceptual characteristics of vibration-augmented buttons, Park et al. [36] recently performed an in-depth analysis using a perceptual space. Their findings are summarized in Section 3.

2.2 Information Transfer of Vibrotactile Stimuli

Humans have a limited ability to identify different sensory stimuli. The information-theoretic framework is a fruitful approach to studying the limited sensory capacity. Here, the information transfer (IT) means the amount of information obtained by communication and is equivalent to the amount of reduced uncertainty [40]. It is beneficial for characterizing performance when the task involves the correct identification of one stimulus from a set of alternatives [52]. The IT measure is usually independent of the task context and thus allows the information transmission capabilities through different sensory channels and displays to be quantified and compared [50]. Achieving high information transfer requires that the stimuli be distinguished reliably from each other.

In this work, we diversify the fixed haptic response of a physical button by superimposing different vibrations. Hence, the information transmission capacity of vibrotactile stimuli is pertinent to our research. Researchers have measured the IT achievable with vibrotactile stimuli in various conditions [4, 8, 45, 49, 51, 57]. The design variables tested include frequency, amplitude, duration, number of pulses, and spatial location.

Early research used only single-pulse vibrations. Summers et al. [49] reported that IT was 1 bit out of the maximum 1.97 bits for vibrations of different frequencies (50 and 100 Hz). Ryu et al. [42] designed vibrations that were equally spaced in perceived intensity on a mobile device. The measured IT was 1.68 bits (maximum 2.32 bits). Chen et al. [9] designed virtual buttons with only vibration feedback. They performed user experiments to find clearly recognized patterns and found that 5 to 6 virtual buttons were perfectly identified among 7 virtual buttons (2.5 bits out of 2.8 bits). Sherrick [45] confirmed the important effects of vibration frequency and amplitude on IT while increasing the frequency from low (2 Hz) to high (290 Hz). The IT score was 2.23 bits when only the frequency was varied and 2.67 bits when both the frequency and amplitude were changed (maximum 3.32 bits). Recently, Yoo et al. [57] measured IT values using a large number (90) of vibration patterns with different numbers of pulses, tempos, frequencies, and amplitudes. When only one pulse was used, the IT score was 1.31 bits out of 4.16 bits. IT

³Virtual buttons can also be made by altering the force-displacement response [29, 53]. This approach can lead to highly realistic feedback, but the system is generally bulky and costly. For the context of our research, we confine the discussion in this paper to the more popular and straightforward approach of vibrotactile feedback.

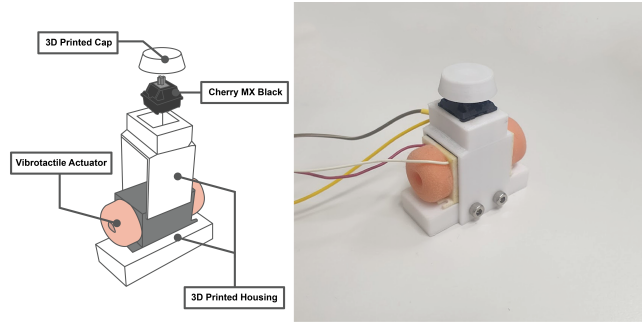


Figure 2: Augmented button replicated from [36]. The button is 51 mm × 25 mm × 55 mm in size and 55 g in weight. The button cap has 17.5 mm of contact diameter and 6.6 mm of height.

of vibrotactile stimuli was also measured for a tablet [38], a wrist band [8, 10, 24, 26, 28], and rings [37].

A dedicated review of the information-theoretic framework and a more complete survey of haptics-related results are available in Tan et al. [50]. To our knowledge, the amount of information transferred by augmented buttons has not been quantified.

3 VIBRATION-AUGMENTED BUTTONS

Our design and selection of vibration-augmented buttons were inspired by the previous study of Park et al. [36]. Thus, we replicated their hardware (Figure 2) and used it in user experiments. The augmented button used a linear mechanical switch (Cherry MX Black) exhibiting no buckling effects and a voice-coil vibration actuator (Haptuator BM3C, Tactile Labs) of a wide frequency bandwidth.

Park et al. [36] augmented the linear physical button with 40 vibrations combining four frequencies (40, 100, 250, and 400 Hz), two amplitude levels (full and half), and five envelopes. They also included seven physical buttons of different responses as references. The perceptual distances between each pair of the 47 physical and augmented buttons were estimated by a user experiment. A perceptual space that best preserved the perceptual distances had two dimensions. The significant physical variables were vibration frequency and energy level. The envelope's shape and duration determined the energy level, and it had three levels (low, medium, and high). The amplitude level was insignificant.

Based on these results, we decided to consider only 20 augmented buttons, shown in Figure 3 (left), by fixing the amplitude level to full. The 20 augmented buttons were categorized by vibration frequency and energy level.

The study of Park et al. [36] also included an experiment for adjective rating. Each augmented button was rated against seven adjective pairs of *sharp-blunt*, *even-bumpy*, *light-heavy*, *smooth-sticky*, *soft-hard*, *shallow-deep*, and *cheap-luxurious*. The best perceptual dimensions spanning the 2D perceptual space of the 47 buttons were *sharp-blunt* and *soft-hard*.

The supplemental materials in Park et al. [36] provide an affective space that presents the point coordinates for the two perceptual dimensions of *sharp-blunt* and *hard-soft*. Such an affective space is adequate for representing or synthesizing haptic sensations [15].

The affective space is shown in Figure 3 (right). A numbered point represents the sensation of the corresponding button in the form of (x, y) , where x and y are the coordinates in the *sharp-blunt* and *hard-soft* axes, respectively. Button #0 represents the base physical button without augmentation. #1–#20 are augmented buttons using the same-numbered vibrations shown in Figure 3 (left). B1–B7 are physical buttons with different force-displacement responses and are shown for reference; also see Section 5 and Figure 9. The distance between two points corresponds to the perceptual dissimilarity between the two corresponding buttons. The affective space indicates that the feel of a general physical button can be altered to different sensations by varying the frequency and envelope of the superimposed vibration.

The affective space provides valuable information for composing a set of augmented buttons. For example, including augmented buttons far from each other in the affective space can improve the set's information transmission capacity. When mapping augmented buttons to abstract meanings for application design, the perceptual quality of an augmented button should agree to the assigned meaning to facilitate users' learning and recall. The perceptual quality is expressed by the degrees in *sharp-blunt* and *hard-soft* in the affective space. The regression results of other adjective pairs, such as *even-bumpy*, *light-heavy*, *smooth-sticky*, *shallow-deep*, and *cheap-luxurious*, are also available in [36, Figure 11]. We selected augmented buttons to use in our research using the affective space.

4 INFORMATION TRANSMISSION CAPACITY OF AUGMENTED BUTTONS

The information transmission capacity determines the recognition ability of the human tactile system regarding augmented buttons: *how many augmented buttons users can perfectly identify*. To estimate the information transmission capacity, we measured the information transfer of augmented buttons in an absolute identification experiment while increasing the number N of augmented buttons from 4 to 10. For each N , we selected augmented buttons based on their affective space described in Section 3.

4.1 Methods

This experiment and the other ones reported in the next sections were all approved by the institutional review board at the authors' institution (PIRB-2021-E029).

4.1.1 Experimental Conditions. The experiment measured the ITs of four sets of augmented buttons with $N = 4, 6, 8$, and 10. For each set, we tried to select the most distinguishable augmented buttons. Our strategy was straightforward: (1) Choose the same or similar numbers of augmented buttons from the four quadrants in the affective space so that their perceptual distances will be the greatest and (2) Balance other physical properties, such as frequency, energy, and duration (vibration duration is a critical factor for discrimination and identification). The selected augmented buttons are shown in Table 1. These buttons are coded in different colors in Figure 3. The base physical button #0 with no vibratory effect was always included as a representative of quadrant 2. It is most clearly different from other augmented buttons with additional vibratory sensations.

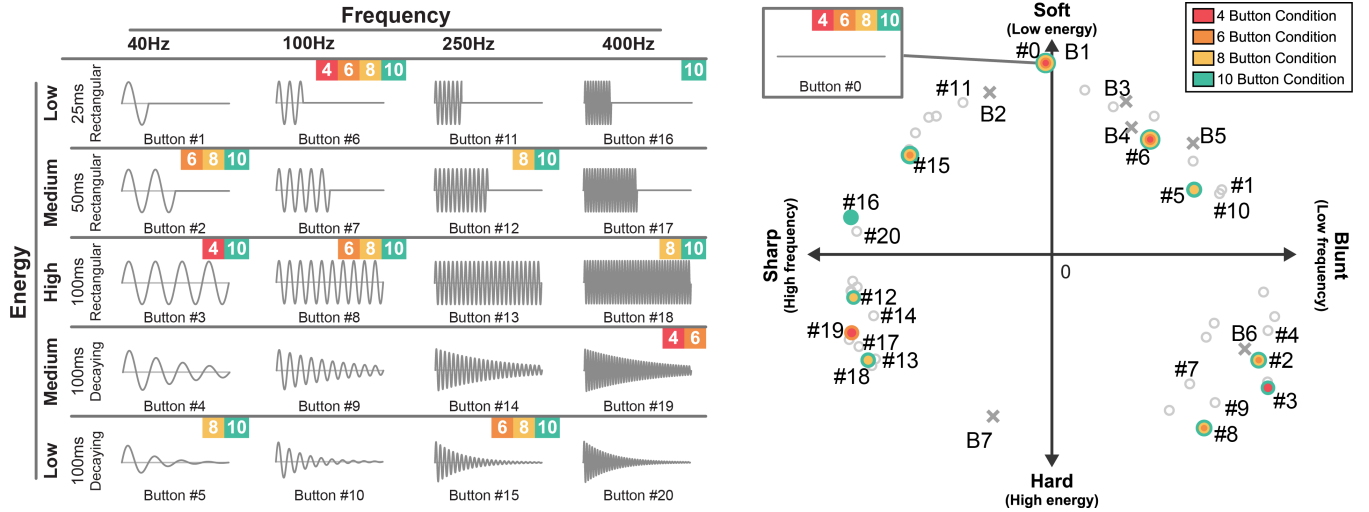


Figure 3: (Left) 20 vibration signals used for button augmentation. The vibrations marked with upper-right color-coded labels were used in the IT estimation experiment. Each number indicates the experimental condition in which the vibration was used; see Section 4 for details. (Right) Affective space of the 20 augmented buttons. Button B1–B7 are physical buttons shown for reference. The same color codes from the left vibration table are used.

Table 1: Augmented buttons selected for IT estimation.

N	Quadrant 1	Quadrant 2	Quadrant 3	Quadrant 4
4	#6	#0	#19	#3
6	#6	#0, #15	#19	#8, #2
8	#5, #6	#0, #15	#12, #18	#8, #2
10	#5, #6	#0, #15, #16	#12, #18	#8, #3, #2

* These buttons can be traced counterclockwise in Figure 3.

4.1.2 Participants. Forty participants (24 males and 16 females; age $M = 23.9$ years and $SD = 1.91$ years) participated in this experiment. Ten participants performed an absolute identification procedure to estimate the IT of each of the four augmented button sets. No participants reported any known sensorimotor abnormalities. They were paid approximately 9 USD per hour as compensation. Before the experiment, participants were informed of the experiment’s goals and procedure via a written document, and then they signed a consent form.

4.1.3 Procedure. The augmented button hardware was placed on a table and enclosed by a black cardboard box (Figure 4, left). Participants sat in a chair in front of the table and placed their left hands to operate the augmented button (Figure 4, right). They had no visual access to the augmented button hardware. Participants used their right hands to control the experiment program using a mouse. They wore noise-canceling headphones that played white noise to block the vibration sound.

Each identification procedure consisted of three sessions: training, practice, and main session. In the training session, participants experienced the N augmented buttons of the assigned set and made their identification criteria for at least 10 minutes. Participants were

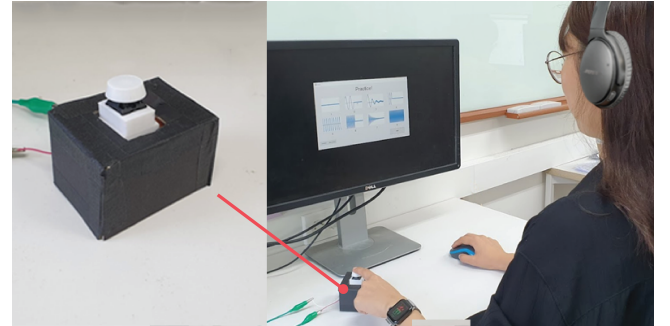


Figure 4: Experimental setup. (Left) The augmented button enclosed by a cardboard. (Right) A participant conducting the IT estimation experiment of vibration-augmented buttons.

provided with a GUI that displayed the vibration signals on the monitor screen (Figure 5). They selected a vibration signal by clicking the signal icon using the mouse. Then, the corresponding vibration was played back when participants pressed the augmented button. Participants wrote down subjective descriptions, such as the characteristics of each button or differences between the buttons’ sensations, on a sheet of paper to establish and assist their identification criteria.

In the practice session, participants were asked to find the correct button among the N augmented buttons. In each trial, one randomly selected augmented button was presented. Participants pressed the button as many times as they wanted to and entered which button it was by clicking the corresponding signal icon on the GUI using the mouse. After each response, correct answer feedback was visually provided in a text box on the GUI. Participants were free to modify

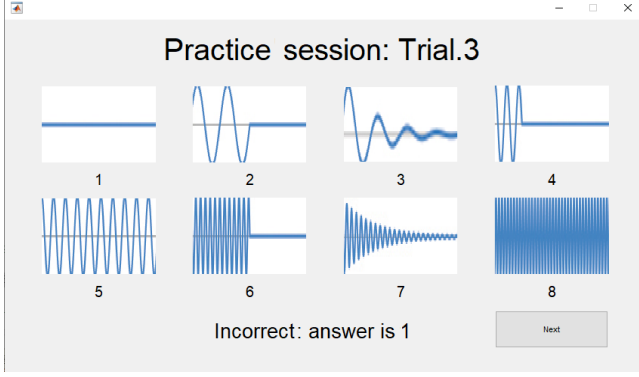


Figure 5: Example GUI showing vibration patterns of augmented buttons ($N = 8$). Correct answer feedback was provided with a text.

and rewrite their memo sheet made in the training session. The practice session consisted of $5N$ trials, e.g., 40 trials for the eight-button set.

In the main session, participants' task was the same as in the practice session, but there was no correct answer feedback. The numbers of trials were 80, 180, and 320 for the 4, 6, and 8 augmented buttons with the repetitions of 20, 30, and 40, respectively. These choices were based on Miller's recommendation [33] that the number of trials should be larger than $5N^2$ to avoid an overestimation of IT. The ten-button set required at least 500 trials to follow the $5N^2$ guideline. However, such many trials made the experiment time unbearably long, and we set the number of trials to 300 with 30 repetitions per augmented button. The main session was divided into five sub-sessions. Participants had a break between sub-sessions to prevent fatigue. The experiment took 30 min, 1 h, 1 h 30 min, and 1 h 30 min on average for the four augmented button sets, respectively.

4.1.4 Data Analysis. We pooled all participants' data from the same set and obtained four $N \times N$ stimulus-response confusion matrices. Then, we computed two metrics: the percent-correct (PC) score, which represents identification accuracy, and the information transfer. The maximum likelihood estimate of IT, IT_{est} was computed by the following standard formula [50]:

$$IT_{est} = \sum_{j=1}^N \sum_{i=1}^N \frac{n_{ij}}{n} \log_2 \left(\frac{n_{ij} \cdot n}{n_i \cdot n_j} \right), \quad (1)$$

where n is the total number of trials, n_{ij} is the number of responses for j to stimulus i , and n_i and n_j are the row and column sums, respectively. $2^{IT_{est}}$ is interpreted as the number of buttons that can be correctly identified without errors.

4.2 Results

The confusion matrices obtained in the experiment are shown in Figure 6 for the four sets of augmented buttons. The identification accuracy was perfect in the four-button set, and it was degraded as the number of buttons increased as expected. The PC scores

were 100% (SD=0%), 92.9% (SD=7.9%), 90.3% (SD=7.7%), and 72.3% (SD=7.6%) for $N = 4, 6, 8$ and 10 .⁴

The estimated IT values are presented in Figure 7 where the maximum achievable values of IT, $\log_2 N$, are also shown side by side for comparisons. IT_{est} was 2 bits (SD=0 bits) for $N = 4$, 2.30 bits (SD=0.24 bits) for $N = 6$, 2.60 bits (SD=0.25 bits) for $N = 8$, and 2.42 bits (SD=0.17 bits) for $N = 10$. The IT_{est} values were saturated with increasing N , indicating that the information transmission capacity of vibration-augmented buttons is approximately 2.60 bits. The numbers of augmented buttons that can be perfectly recognized, $2^{IT_{est}}$, are 4, 4.94, 6, and 5.35, respectively.

We conducted a one-way between-subjects ANOVA on the IT score for one independent variable of the number N of augmented buttons. N had a significant effect on IT_{est} ($F(3, 36) = 16.81, p < 0.0001$). We also performed Tukey's HSD post-hoc analysis. IT scores significantly increased with N : $N = 4$ vs. 6 ($p < 0.01$) and $N = 6$ vs. 8 ($p < 0.01$). However, the increasing trend was saturated with the greatest N : $N = 6$ vs. 10 ($p = 0.56$) and $N = 8$ vs. 10 ($p = 0.18$).

In conclusion, the IT score was maximized to 2.6 bits when the eight augmented buttons were provided and then plateaued with the inclusion of more buttons. Alternatively, the participants near-perfectly recognized 5 to 6 augmented buttons when more than eight buttons were presented.

4.3 Discussion

4.3.1 Comparisons Between Buttons Sets. In the set of four augmented buttons, all the four buttons were recognized perfectly. The four buttons represented the distinct perceptual characteristics of the four quadrants in the affective space shown in Figure 3. Augmented buttons in quadrant 1 feel blunt and soft, those in quadrant 2 sharp and soft, those in quadrant 3 sharp and hard, and those in quadrant 4 blunt and hard. Besides, all the four buttons had different vibration frequencies (none, 40, 100, and 400 Hz), which can increase the IT [5, 8, 57]. They also had different envelope shapes and durations, which makes identification easier [9]. These mutually distinctive perceptual and physical features contributed to the perfect identification of the four buttons.

In the six-button set, the PC score and IT_{est} were 92.9% and 2.3 bits, respectively. The participants could identify approximately five augmented buttons perfectly among the six buttons. Eight participants identified most of the buttons correctly. However, two participants showed considerably lower IT_{est} values: 1.93 and 1.92 bits, respectively, making almost all errors observed with this button set. The two participants occasionally confused buttons #8 and #19 and buttons #15 and #19; see the confusion matrix in Figure 6. The vibrations used for the three buttons #8, #15, and #19 had the same duration (100 ms), while the others had shorter durations (Figure 3).

IT_{est} was maximized to 2.6 bits, which corresponds to six perfectly identifiable items, in the eight-button set. The off-diagonal terms in the confusion matrix were all very low, except for two cases: button #6 was misrecognized as #5 (14.2%) and button #12 as #15 (16.0%). Button #6 and #5 had relatively similar frequencies (100

⁴According to Tan et al. [50], the percent-correct score in an absolute identification task is not an ideal performance measure as it drops as the task difficulty increases. In contrast, IT plateaus and provides an unequivocal measure of human performance as the channel capacity.

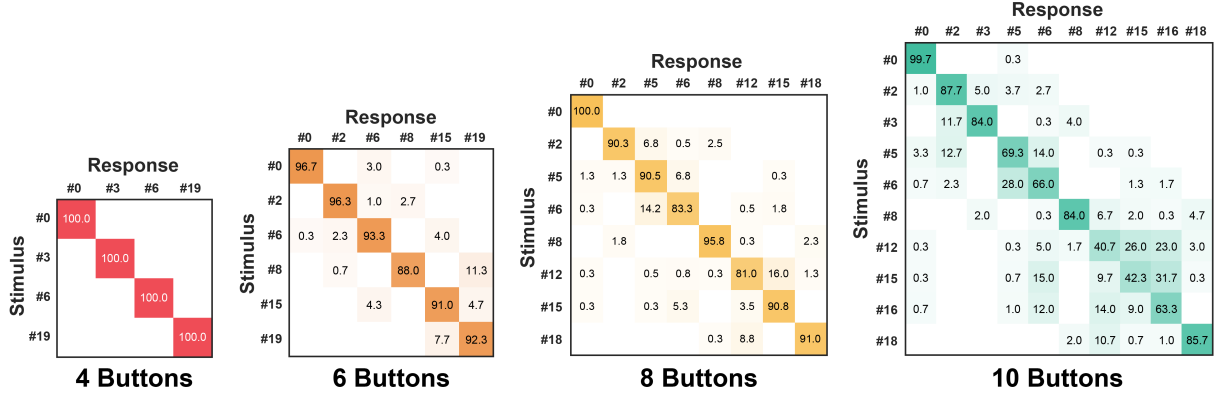


Figure 6: Confusion matrices for the four button sets.

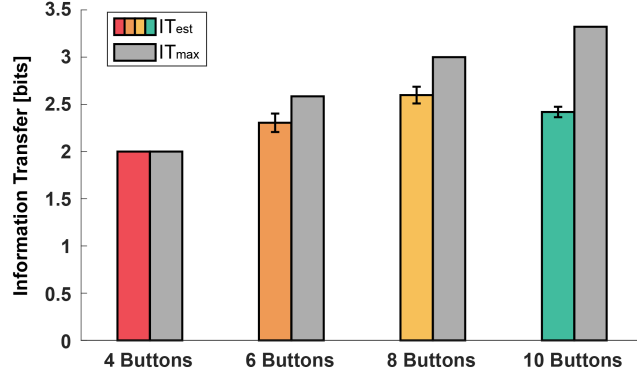


Figure 7: Estimated IT values (colored bars) and maximum IT values (gray bars). Error bars represent standard errors.

and 40 Hz). However, button #6 had a short duration (25 ms) and a rectangular envelope, and button #5 with a long duration (100 ms) had a fast-decaying envelope; see Figure 3. Thus, the actually-generated vibrations could be more similar than the input signals between the two.⁵ A similar observation is applied to button #12 and #15.

In the last ten-button set, $IT_{est} = 2.42$ bits, and it was similar to but slightly decreased from the IT_{est} of the eight-button set without statistical significance. The confusion matrix in Figure 6 shows that the two newly-added buttons #3 and #16 caused substantial confusions to buttons #2, #3, #5, and #6 and buttons #12, #15, and #16, respectively. In general, a curve of IT for the increasing number of alternatives shows a plateau after initial increases [50, 52]. Therefore, we determined that $N = 10$ reached a plateau and stopped testing higher N .

We conclude that the human channel capacity is bounded by 2.6 bits for the vibration-augmented buttons selected by our design strategy. Therefore, we expect users to recognize as many as 5–6 augmented buttons with no errors. This information-theoretic

⁵Every vibration actuator makes a residual vibration. It makes the output for an input signal of a rectangular envelope longer than the input signal's duration.

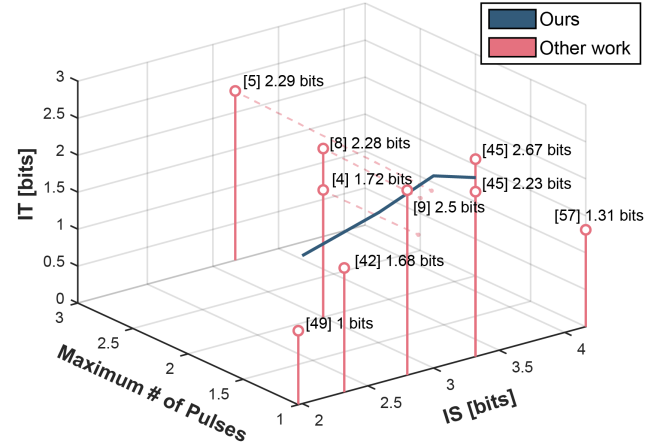


Figure 8: IT values reported in previous studies (pink lines and dots) and our IT values (blue line).

analysis provides a context-independent guideline on the number of augmented buttons usable for interaction design.

4.3.2 Comparisons with Previous Studies. Vibration-augmented buttons add programmable vibrations to the constant kinesthetic response of a physical button. Hence, it is probable that the IT of augmented buttons is lower than the IT of vibrotactile stimuli only; the button's kinesthetic response might mask the subtle changes in vibration to some extent. For comparison, we surveyed the IT values of vibrotactile stimuli applied to a single location on the fingertip from the literature.⁶ They are shown in Figure 8 as a function of the information in stimuli (IS) and the maximum number of pulses used in vibration pattern design; also see Section 2.2 for a detailed review. The highest IT of 2.6 bits of our augmented buttons is comparable to or better than the IT values of single-pulse vibrotactile stimuli reported in the literature. This high IT is despite the co-present

⁶Vibrotactile stimulation to multiple body locations significantly improves information transfer. For example, the IT values reported were 4 bits for two locations [41], 4.28 bits for three locations [28], and as high as 7.02 bits for four locations [37]. Refer to Tan et al. [50] for a detailed survey.

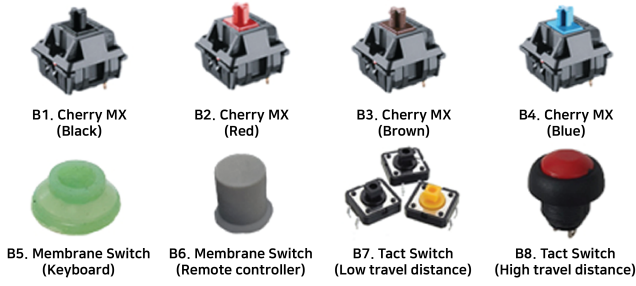


Figure 9: Eight physical buttons.

kinesthetic stimuli, presenting evidence for the effectiveness of our vibration-augmented button design using the affective space.

We expect that using multiple pulses of vibrations can improve the IT capacity further. However, the literature has not presented sufficient evidence. Azadi and Jones [4, 5] evaluated IT on the forearm and fingertip by varying frequency, amplitude, and also the number of pulses, and the IT score was 2.29 bits. The IT measured by Chase et al. [8] on the forearm using two-pulse vibrations was 2.28 bits. Exploring a design space of multiple-pulse vibrations to increase the IT of augmented buttons can be an interesting topic for future research.

5 INFORMATION TRANSFER OF PHYSICAL BUTTONS

After studying the information transmission capacity of augmented buttons, we had a similar question for physical buttons. We carried out another absolute identification experiment to measure the IT of commercial physical buttons that had different operating structures and feels. This result can be compared to the IT values of augmented buttons reported in the previous section to judge the adequacy of our augmented button design.

5.1 Methods

5.1.1 Stimuli and Hardware. This experiment used eight physical buttons based on the result that IT was maximized at eight vibration-augmented buttons reported in Section 4. Figure 9 shows the eight physical buttons. Buttons B1–B7 were the same ones used in [36], exhibiting diverse perceptual characteristics as shown in Figure 3. Button B8 was selected because of its distinctive feel.

The eight physical buttons had different characteristics. Buttons B1–B4 were Cherry MX switches used for mechanical keyboards. B1 and B2 were linear switches that did not make buckling effects. They had different operating forces for the keypress. B3 and B4 showed nonlinear responses exhibiting buckling effects. They also required different operating forces. Detailed data sheets and force-displacement curves are available in [11] for B1–B4. B5 and B6 were membrane switches. B5 was taken from a keyboard, and B6 was taken from a remote controller. B6 had a higher stiffness than B5. B7 and B8 were tact switches that present strong and clear buckling effects widely used for electronic devices and cars. B8 had a larger travel distance than B7.

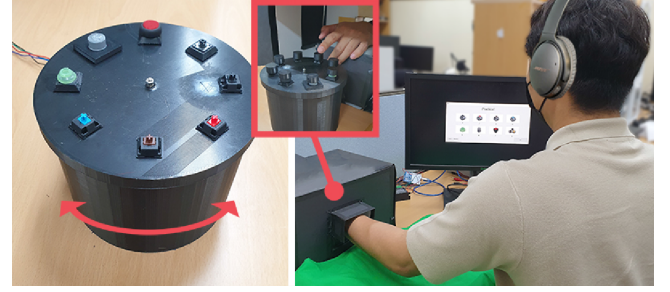


Figure 10: Experiment setup (right) with a housing that automatically rotates eight physical buttons with caps removed (left) and wearing the same caps (inset).

We built a rotational 3D-printed housing (radius 160 mm and height 105 mm) shown in Figure 10 (left) for automatic positioning of the eight physical buttons. The eight buttons were uniformly placed on the top cover. A step motor (Nema17 42, STEPPERON-LINE) in the housing rotated the cover. The same 3D-printed round caps were placed on all the physical buttons to prevent the tactile cues provided by the original caps of different shapes. The hardware system was controlled by a PC.

5.1.2 Participants. Ten volunteers (6 males and 4 females: age $M = 24.2$ years and $SD = 1.68$ years) participated in this experiment. No participants reported known sensorimotor disorders. Before the experiment, participants were informed of the experimental procedure and signed a consent form. They were paid approximately 9 USD after the experiment.

5.1.3 Procedure. The button housing was enclosed by a large black cardboard box to block visual cues and placed on a table. Participants put their left hands in the box and pressed a button with a finger. The button to press was selected by the experiment program and placed automatically under the finger by rotating the housing's top cover. A silicon pad was placed beneath participants' wrists to make the posture comfortable.

The experiment procedure was similar to that for augmented buttons. Participants were provided with a GUI showing the pictures of the eight physical buttons. The experiment consisted of a training session of 5 min, a practice session (40 trials of 8 buttons $\times$ 5 repetitions) with correct answer feedback, and the main session (192 trials of 8 buttons $\times$ 24 repetitions) without correct answer feedback. The experiment took 50 min on average to complete.

5.2 Results and Discussion

The confusion matrix obtained from the experiment is shown in Figure 11. The PC score was 74.7% ($SD = 9.64\%$). The IT estimate was 2.32 bits ($SD = 0.33$ bits), or equivalently, five button click sensations could be identified with no errors.

The participants identified the four membrane and tact switches (B5–B8) very well. The four buttons' different buckling sensations, travel distances, and operating mechanisms seem to have contributed to the near-perfect performance. However, the four Cherry MX switches (B1–B4) showed considerably lower recognition performance. Despite their different operating forces, distinguishing

	Response							
	B1	B2	B3	B4	B5	B6	B7	B8
B1	50.4	24.6	21.7	3.3				
B2	36.3	40.0	20.4	3.3				
B3	7.1	27.5	48.3	16.3			0.8	
B4	1.3	2.5	12.9	78.3	2.1		2.9	
B5	0.8				95.0	4.2		
B6					1.3	95.8		2.9
B7	0.4		0.8	1.7		3.8	92.5	0.8
B8						0.4	2.1	97.5

Figure 11: Confusion matrix of eight physical buttons.

between the two linear buttons (B1 and B2) was quite difficult. B2 and B3 were also difficult to discriminate due to their similar operating forces despite B3's buckling effect. B4, which had the highest operating force (60 cN) and a unique buckling feel, was identified better than the other Cherry MX switches.

In our experimental results, the IT of the physical buttons was slightly lower than the IT of the vibration-augmented buttons. However, this is not a definite result because testing with more diverse physical buttons has a chance to demonstrate higher IT values. It would be essentially determined by the ITs between kinesthetic stimuli (physical buttons) vs. vibrotactile stimuli (augmented buttons). Nonetheless, the IT comparison provides another support for the value of vibration-augmented buttons as a means to present distinct key-click sensations to users by adding a simple vibration actuator to a plain physical switch.

6 APPLICATION: VR TEXT ENTRY

In principle, vibration-augmented buttons provide feedback after an action of pressing a key. To showcase their anticipated benefits, we designed and implemented an application for typing texts using a handheld controller in a virtual environment. Using 3D printing, we made a mock-up VR controller that included one augmented button and an HTC Vive tracker (Figure 12). A user wearing a head-mounted display (HMD) enters texts by pointing the controller to the letters on a standard visual QWERTY keyboard displayed on the HMD screen (Figure 13) and then pressing the button. The tracker measured the position and orientation (pose) of the controller. A virtual key selected by casting a ray from the controller's pose to the virtual keyboard is visually highlighted. This is a typical text entry method provided with commercial HMDs. Its performance is significantly better than other text entry methods for VR [48]. Here, we add a simple extension: when pressing the augmented button to enter letters, the user is notified of several confirmation messages encoded in different key-click sensations.

Normally, physical buttons in a VR controller always provide the same haptic feedback.⁷ In contrast, our augmented button enables variable confirmation feedback. For simplicity, we assume that the VR system is intelligent to detect whether the user has entered an

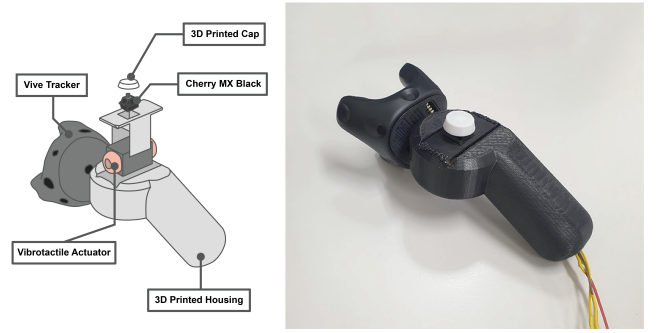


Figure 12: VR controller with a vibration-augmented button.



Figure 13: Virtual environment for text entry using a visual QWERTY keyboard.

acceptable letter or made an error by using a spell checker or a grammar engine. As an exemplar design, we consider six messages that the VR system can provide after a key input and assign six augmented buttons to them.

The design is realized in a finite state machine shown in Figure 14. The state machine consists of three states and six system messages (output). We limited the number of system messages for error feedback to six because it was the upper bound of perfectly-recognizable augmented buttons (Section 4). To choose the system messages, we observed the errors occurring during normal use of the VR typing interface (with no error feedback) and selected the most frequent error types.

The specific mappings from the six system messages to six augmented buttons are as follows:

- (1) When the system message is Correct (no error), we feedback the augmented button #5. This button presents a soft and blunt, even pleasing sensation.
- (2) The system message can be Spelling Error, and this case is notified by button #18. Similarly, Capital Letter Error issues button #12, and Spacing Error triggers button #15. These three buttons feel sharp (Figure 3) and could be identified accurately (Figure 6). We expected that Spelling Error would

⁷Some virtual typing systems available for commercial HMDs issue different sounds for the keypresses leading to system state changes. This work does not consider the potential interplay between variable sound and haptic feedback for simplicity and focuses on haptic feedback only.

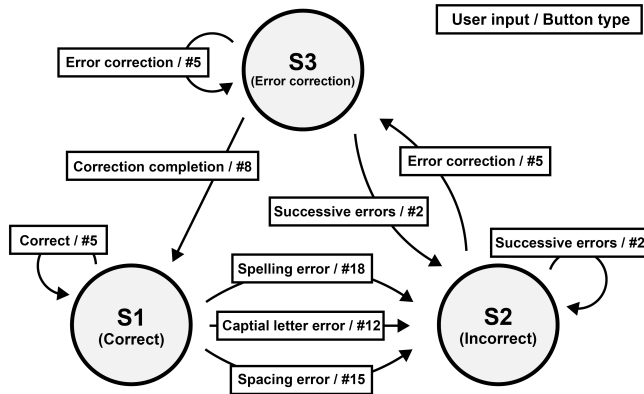


Figure 14: State machine for our VR text entry system using vibration-augmented buttons. Each transition arrow is labeled with the system message (input) and the augmented button (output).

be the most frequent and assigned it with button #18 that had the highest energy among the three buttons.

- (3) A user can make Successive Errors without correcting previous errors. In this case, the system provides haptic feedback using button #2. This augmented button uses a low-frequency and medium-energy vibration, presenting a blunt and hard sensation.
- (4) When all errors are corrected, the system renders button #8. This augmented button feels most different from the other buttons with the highest identifiability (Figure 6).

We also confirmed the adequacy of the above design by conducting a pilot study.

7 USER STUDY: VR TEXT ENTRY

Lastly, we conducted a user study to evaluate the benefits of our VR text entry application that uses a vibration-augmented button. This user study compared two conditions: providing constant or variable haptic feedback at keypress.

7.1 Methods

7.1.1 Participants. Twenty participants (12 males and 8 females; age $M = 23.2$ years and $SD = 3.2$ years) took part in this user study. All participants were right-handed and did not report sensorimotor abnormalities. They were informed of the experiment procedure and signed a consent form before starting the experiment. They were paid approximately USD 9 after the experiment.

7.1.2 Task. The VE consisted of a virtual standard QWERTY keyboard, a transcribed phrase, and a text input box (Figure 13). All of them had fixed positions and orientations. Participants wore a HMD (HTC VIVE Pro) and interacted with the virtual keyboard using the augmented-button controller and the pointing method, as described in Section 6. Participants' task was to type the phrase displayed on the screen as quickly and accurately as possible. Errors were corrected using the delete key (working as the backspace key

on Microsoft Windows). Sixteen transcribed phrases also containing capital letters were chosen from the 500 phrases in MacKenzie and Soukoreff [31]; see Table. 2. The VE was implemented using the Unity game engine.

Table 2: Phrases used for VR text-entry user study.

can see the rings on Saturn	beware the ides of March
February has an extra day	a tumor is OK provided it is benign
I took the rover from the shop	Mary had a little lamb
Canada has ten provinces	I listen to the tape everyday
I am wearing a tie and a jacket	where did I leave my glasses
can I skate with sister today	the collapse of the Roman empire
play it again Sam	do you like to shop on Sunday
I just cannot figure this out	I like baroque and classical music

7.1.3 Experimental Conditions. During text entry, haptic feedback was provided to participants in two conditions. In one condition, *Multiple Patterns (MP)*, haptic feedback used the six augmented buttons depending on the system message, as described in Section 6. In the other condition, *Single Pattern (SP)*, haptic feedback was always the same regardless of the system message using the augmented button #5 (for Correct in the *MP* condition). The *SP* condition was the reference representing the current practice.

7.1.4 Procedure. Participants sat in a chair wearing the HMD. They entered the phrases displayed on the screen by operating the VR controller with their right hands. They wore headphones that played white noise to block any sound effects.

In the training session, participants familiarized themselves with the VR system and experienced text entry in both experimental conditions. They entered two phrases under the *SP* condition and another two phrases under the *MP* condition. The experimenter explained only the entry method of pointing and selection using the VR controller to participants. The experimenter only mentioned that sometimes button sensations could be different but did not explain any differences in haptic feedback between the two conditions.

The main session consisted of four sub-sessions. In each sub-session, participants entered four phrases under one experimental condition and then another four phrases under the other condition. This design was expected to reduce the order effect further compared to an alternative of finishing all the 16 phrases under one condition and then doing the same under the other condition. Participants could proceed to the next phrase only after one phrase was entered correctly. The phrases were randomly ordered per experimental condition and participant. Half of the participants entered the phrases under the *SP* condition first, and the other half did so under the *MP* condition first, for order balancing. Participants rested for 30 s between the experimental conditions while wearing the HMD and 1 min between sub-sessions after taking off the HMD. After the last sub-session, participants filled out a questionnaire on cognitive workload and haptic user experiences. The experiment took approximately 50 min to complete.

7.1.5 Performance Metrics. To evaluate task performance, we measured text input speed and accuracy by calculating the words per minute (WPM) and the error rate, respectively, across the 16 phrases

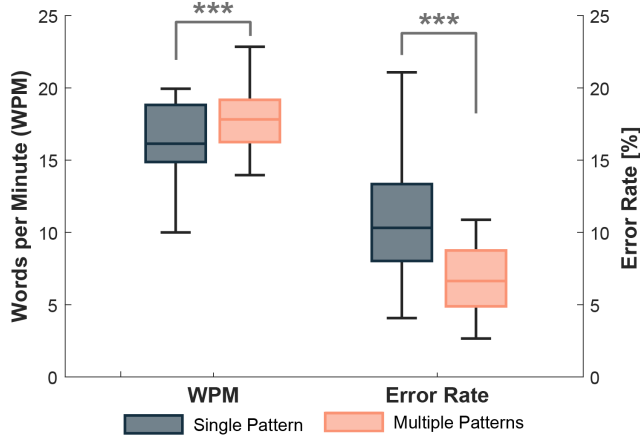


Figure 15: Words per minutes (WPM) and error rates.

using the standard formulas [55]. The WPM was calculated as:

$$WPM = \frac{|T| - 1}{S} \times 60 \times \frac{1}{5}, \quad (2)$$

where $|T|$ is the length of the phrase entered by the participant, and S is the time between the first and last keys (in seconds). The error rate was computed by:

$$Error\ Rate = \frac{IF}{C + IF} \times 100, \quad (3)$$

where IF (incorrect-fixed) is the number of characters backspaced during text entry, and C is the number of all characters in the transcribed phrase.

For cognitive workload, we used the standard NASA-TLX questionnaire using all six factors (mental demand, physical demand, temporal demand, performance, effort, and frustration level) [14]. We also collected data for user experiences by referring to a recent questionnaire specialized in haptic experiences [20]. We included two questions for pragmatic experiences: (Utility) “Is it useful?” and (Causality) “Can you identify the cause of feedback?” and another two questions for hedonic experiences: (Expressivity) “Is there a distinction between haptic effects?” and (Autotelics) “Does it feel good in and of itself?”. All questions were answered on a continuous scale between 0 and 100 on pieces of paper.

7.2 Results

The results of task performance are depicted in Figure 15. The average WPMs were 16.48 ($SD = 2.51$) for the *SP* condition and 18.10 ($SD = 2.35$) for the *MP* condition. The average error rates were 10.92% ($SD = 4.24$) and 6.75% ($SD = 2.44$), respectively. A one-way repeated-measures ANOVA indicated that the haptic feedback condition had a significant effect on WPM ($F(1, 19) = 28.65, p < 0.001, \eta^2 = 0.60$) and error rate ($F(1, 19) = 30.32, p < 0.001, \eta^2 = 0.61$). Therefore, the variable haptic feedback of *MP* improved the text entry performance in VR than the standard constant haptic feedback of *SP*.

Figure 16 (left) shows the results as to cognitive workload. The average weighted score representing the overall workload was 59.93 ($SD = 12.77$) for *SP* and 40.95 ($SD = 15.22$) for *MP*. A significant

difference was found in the weighted scores by one-way repeated-measures ANOVA ($F(1, 19) = 26.13, p < 0.001, \eta^2 = 0.58$). This tendency was unanimous in all individual workload metrics: mental demand ($F(1, 19) = 24.06, p < 0.001, \eta^2 = 0.56$), physical demand ($F(1, 19) = 6.67, p = 0.018, \eta^2 = 0.26$), temporal demand ($F(1, 19) = 13.08, p = 0.0018, \eta^2 = 0.41$), performance ($F(1, 19) = 8.34, p = 0.0094, \eta^2 = 0.27$), effort ($F(1, 19) = 53.30, p < 0.001, \eta^2 = 0.74$), and frustration ($F(1, 19) = 7.96, p = 0.0109, \eta^2 = 0.27$). Therefore, the variable haptic feedback demanded a lower workload than the constant haptic feedback.

The results on haptic user experiences are shown in Figure 16 (right). The average scores of *MP* were higher than those of *SP* for all four questions. The differences were significant in three questions: utility ($F(1, 19) = 27.61, p < 0.001, \eta^2 = 0.54$), causality ($F(1, 19) = 27.80, p < 0.001, \eta^2 = 0.59$), and expressivity ($F(1, 19) = 30.25, p < 0.001, \eta^2 = 0.59$), but not for autotelics ($F(1, 19) = 1.7, p = 0.208, \eta^2 = 0.07$).

7.3 Discussion

This user study demonstrated how to utilize vibration-augmented buttons in VR scenarios. The event-driven variable haptic feedback enabled by one augmented button presenting multiple haptic sensations improved the VR text transcription task in all aspects of task performance, cognitive workload, and user experience. This result is the first empirical evidence for the user benefits of augmenting physical buttons with additional haptic feedback.

7.3.1 Task Performance. Our implementation of the *SP* condition resulted in the average WPM of 16.48, and this value is comparable to the WPMs of other QWERTY-based pointing methods: 13.2 in Markussen et al. [32], 18.9 in Shoemaker et al. [47], and 15.5 in Speicher et al. [48]. Compared to the *SP* method, our *MP* method improved the typing speed by approximately 2 WPM (9.8%) and reduced the error rate by 38%.

To obtain more insights, we counted the numbers of the four kinds of errors recorded during the experiment. The results are summarized in Table 3. The participants made a total of 1363 errors under *SP* and 704 errors under *MP*. This comparison clearly accounts for the improved performance of *MP*.

Table 3: Error distributions in VR text entry study.

	Single Pattern		Multiple Patterns	
	Count	Perc. (%)	Count	Perc. (%)
Spelling Error	604	44.31	584	82.95
Capital Letter Error	29	2.13	22	3.13
Spacing Error	14	1.03	19	2.70
Successive Error	716	52.53	79	11.22
Sum	1363	100	704	100

In Table 3, the numbers of spelling errors, capital letter errors, and spacing errors were similar between the two conditions. This result is expected because *SP* provides no error feedback, and *MP* makes feedback only after errors are detected. The three errors are the first ones made after a series of correct entries, and no

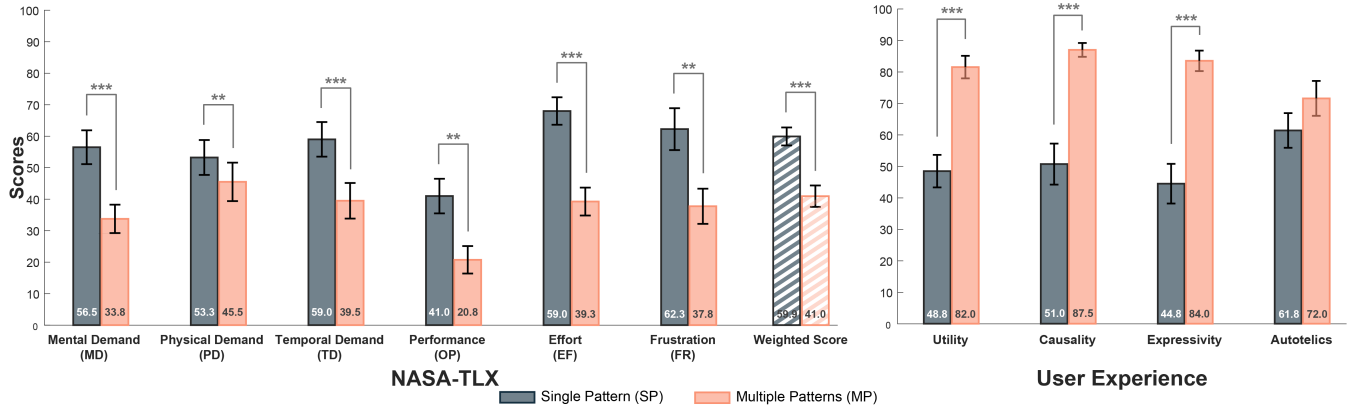


Figure 16: Cognitive workload measured using NASA-TLX (left) and user experience metrics (right). Significant differences are represented with asterisks (: $0.001 < p < 0.01$, *** : $p < 0.001$).**

feedback methods can reduce them (we would need a method for prediction and warning). However, it seems that the participants well recognized the haptic error messages of *MP* and corrected the errors instead of making successive errors. Compared to the 716 successive errors of *SP*, those of *MP* were greatly reduced to 79, only 11% of *SP*.

Regarding the response time, previous studies demonstrated that users respond faster when provided with both visual and tactile feedback than only visual feedback [27, 34]. Our results indicate that the response time can be further improved when tactile feedback is diversified to deliver different meanings pertinent to the task.

7.3.2 Cognitive Workload. In all metrics of the NASA-TLX questionnaire, the event-driven haptic feedback significantly alleviated the cognitive workload imposed on the participants compared to the standard constant haptic feedback. It appears that the participants became aware that the variable haptic feedback was for error correction. Such practical and psychological assistance can bring positive effects on the perceived workload.

7.3.3 Haptic User Experiences. Our text entry method with variable haptic feedback significantly improved the two pragmatic metrics of utility and causality and one hedonic metric of expressivity. The three scores of *MP* were also very high, over 80 out of 100, whereas those of *SP* were below 50.

The positive result for causality is encouraging. We did not explain any of the cause-and-effect relationships of variable haptic feedback to the participants. The participants must have figured the relationships out, at least partially, for themselves during the short training session (and maybe the main session). Such good learnability could have originated from the high expressivity of *MP*.

7.3.4 Limitations. This evaluation study used a transcription task, which corresponds to an ideal situation where the system can perfectly detect all errors. This property makes the task appropriate for isolating the effectiveness of an error feedback scheme in assessment; in our case, the viability of vibration-augmented buttons. However, in more general cases, some functions, e.g., detecting a spelling error for each input of letter, are difficult to implement.

Thus, the performance improvements reported here must be considered as the upper limit achievable with variable haptic feedback.

8 CONCLUSIONS

Vibration-augmented buttons are a promising platform that allows us to deliver different sensations and meanings when users operate physical buttons—a universal interface for almost all devices. This paper presented three fundamental and practical studies with a common goal to deepen our understanding of vibration-augmented buttons. In the first study, we estimated the information transmission capacity of augmented buttons in a large-scale absolute identification experiment. We found that the amount of information that humans can recognize from augmented buttons is limited by 2.6 bits. This result indicates that users can identify as many as 5–6 augmented buttons without confusion; this number is context-free. In the second study, we verified that vibration-augmented and physical buttons are similar in information transmission performance.

The third study demonstrated how to apply vibration-augmented buttons to VR applications using text entry with a handheld controller as the subject. We designed a message-driven variable haptic feedback method to reduce input errors and demonstrated its improved task performance, cognitive workload, and user experience. This result instantiates the value of vibration-augmented buttons as a new tool for providing effective and enjoyable interactions.

We expect our findings to be directly extended for error correction in other text-input interfaces, such as a keyboard for computers. Programmers may find it handy if different keyclick sensations notify syntax errors during programming. Other interaction spaces where vibration-augmented buttons can be beneficial include: (1) gaming where gamers are provided with diverse haptic feedback as the results of their control actions, (2) eye-free interaction where non-visual information is critical [7, 56], and (3) affective interaction where the keyclick feedback is customized to individual users or specific situations.

One research issue that deserves further attention is how to find easy-to-remember semantic mappings from vibration-augmented buttons to intended meanings. Haptic design of signal-to-meaning

providing good learnability and memorability is a crucial issue that has been studied for more than a decade [12, 17, 44]. It can be more so for augmented buttons that can use only short vibrations. This topic is a pressing item for our future work.

ACKNOWLEDGMENTS

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